



Lecture 1: Introduction

Energy Economics of the Green Transition

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Monday 7 September 2026

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1. **The course:** Who, what, how, exam, AI.
2. **Basics:** Why energy and what energy economics is.
3. **Hands on:** Online resources, AI assistant, workshop lectures.

The course

Who we are

Rasmus K. S. Berg

- Assistant professor, Department of Economics.
- Environmental, climate and energy economics; some macro and political economy.
- Started the course in 2022. Lectures 1–7.

Janek B. Eskildsen

- Danish Industries, Green Analysis.
- Formerly Secretariat of the Danish Council on Climate Change.
- Formerly² Postdoc at UCPH.
- Started the course in 2022. Lectures 8–12.

Guest lectures. Frederik R. Øvlisen (Danish Utility Regulator) teaches lecture 2. Christian Dahl Winther (EIFO) talks about investments in wind and P2X (45 minutes 26/10). One or two other guests TBA.

What kind of course this is

Energy economics sits between two fields, and this course draws on both.

Engineering

- Technologies and system costs.
- System models with enormous technical detail.
- Limited scope (no general equilibrium dynamics).
- Centralised *optimisation*.

Economics

- Efficiency, externalities, regulation, welfare.
- Stylised representation of technologies.
- Greater scope (general equilibrium dynamics).
- Decentralised *market outcome*: Behavior and alignment of incentives.

What kind of course this is

An **economics** course. We will not discuss how a lithium-ion battery works, but we need its economically relevant characteristics: what it costs, how flexibly it can run, and what limits deploying it at scale.

References for what we need to know about generation technologies in our course collected in (mostly supplementary) lecture note “Generation Technologies”.

The lectures

	Lecture	Week
1	Introduction	37
2	Energy markets, concepts and market design	38
3	Basic power system models	39
4	Storage and flexibility	40
5	Sector coupling and transmission	41
6	Greenfield investments	43
7	European paths to net zero	44
8	Network effects in green technology adoption	45
9	The energy efficiency gap	46
10	Energy policies for the green transition	47
11	CCS, BECCS, DAC and negative emissions	48
12	Energy and geopolitics	49

Lectures 3–7 are one model growing up, from a single hour of dispatch to Europe under a carbon budget. Lectures 8–12 widen out from the power system.

Passive and active learning

Traditional three hour lectures with passive learning to slides do not work. So, lectures (3–7 at least) follow a **workshop** style roughly as follows:

Before lectures: Read excerpts from lecture note and skim read "workshop" paper(s).

Lectures are three blocks arranged something like this:

1. Introduction (first block only): I introduce setup of the paper (15 minutes).
2. Workshop: You work on questions from worksheet (30 minutes).
3. Solutions: We go through solution together (15 minutes).

After lectures: I'll upload solutions.

How to prepare for a workshop lecture

The worksheet is published before class and names the note pages and the results the day's work uses.

1. **Read the "main text".**

We will primarily work on the workshop paper during lectures, so work through this carefully, e.g. with AI assistant (we'll return to this below).

2. **Skim read the workshop paper.** Note the questions it asks, the model/data it uses, main results/claims. Don't spend hours here.

3. **If interested**, look at supplementary material and see if anything catches your attention (we've collected a lot, so hopefully there'll be something).

If you did not do the reading, come anyway. Each part gives the result it needs at the top; you will be slower, not stuck.

The exam

- **Closed book**, 3 hours at UCPH. No internet, no AI, no programming.
- **Exam curriculum:** Literature from [lecture plan on Absalon](#) and content from lecture slides.
- **Not directly relevant for exam/supporting role:** Literature noted as supplementary reading and anything else, including whatever online materials we may add to Github repository.
- **Exam type:** AI forced us to change exam type, so no previous exams to rely on. Mix of formal model analysis and essay like questions. Examples: Derive a result, read a figure and explain a mechanism at play, argue a policy question.

The curriculum

As mentioned above, the **the curriculum** consists of lecture notes, academic papers, excerpts from textbooks, and lecture slides.

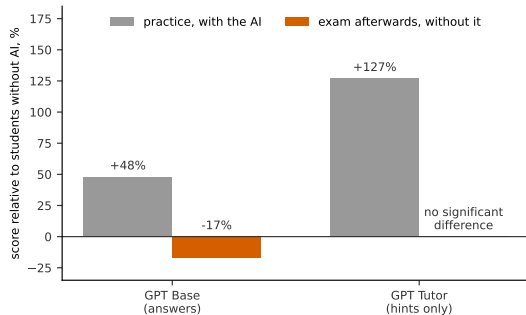
- **Workload:** 1-2 sections from a lecture note and a paper or two (only skimming though) for each lecture.
- **Where:** Lecture plan on Absalon collects it (Github repo holds a copy). Full bibliographic references are collected in separate pdf on Absalon (ReferenceList.pdf).
- **Access:** All materials can be either downloaded directly from Absalon/Github repository (lecture notes) or using [Royal Danish Library](#) with your KU login. Go in through the library, not the publisher's front page.

There is no AI restriction on your preparation, exercises, or anything else during the term. Use whatever you like, as much as you like.

The repository ships an assistant that knows the course material and the model code. It works in GitHub Copilot (free for students) or likely any other setup you work in (e.g. Claude, Mistral) both read the same configuration. Ask it anything, or use its commands:

<i>/hint</i>	one nudge on the problem you are stuck on
<i>/exercise-coach</i>	work through a problem
<i>/model-explainer</i>	what a piece of model code means economically
<i>/check-my-reasoning</i>	critique an argument you have written
<i>/debug-my-setup</i>	installation and import problems
<i>/exam-prep</i>	practice problems and closed-book drilling

The illusory learning effect



Nearly a thousand high-school students, one field experiment.

Bastani et al. (2025), drawn from their Table 1.

Watching a correct solution appear is not the same as being able to produce one. It feels the same.

So the assistant here starts with a question. Ask for the answer and it will give one.

The one request

Turn off Copilot's **inline completions** while you work on exercises. Chat stays on.

Learning takes effort, and that is the point

What you can use later depends on the **processing you did**, not on how many correct answers passed in front of you.

- **Desirable difficulties** (Bjork and Bjork 2011): practice that is harder and slower (testing yourself, generating the answer, spacing) produces more durable learning than practice that is smooth. Retrieving beats rereading (Roediger and Karpicke 2006).
- **Productive failure** (Sinha and Kapur 2021): across 53 studies, students who struggled with a problem *before* seeing the method understood it better and transferred it further.
- The difficulty must be one you can overcome. Floundering is not learning either, which is why each work part gives the result it needs at the top.

Asking an assistant for the answer is the mirror image of all three: useful when checking, costly when learning. And the cost is not only the exam. A skill you did not build is one you do not have.

- Handwriting, not typing, produces widespread connectivity between the brain regions involved in memory and learning (Van der Weel and Van der Meer 2024).
- People who wrote with an assistant could not quote what they had just written (Kosmyna et al. 2025).

In this course, formal analysis and derivations are done by hand.

Absalon

- Lecture plan and dates.
- Reading list.
- Announcements.
- Lecture notes, slides, worksheets.

The repository

github.com/ChampionApe/EnergyEconomics

- Copies of reading list, notes, slides (consistent with Absalon version).
- Models, dataset, notebooks with exercises and notes.
- The AI assistant.

Note: Absalon materials generated from Github repository. Material is added/adjusted during the term; *git pull* keeps repo current and I'll announce adjustments made to Absalon.

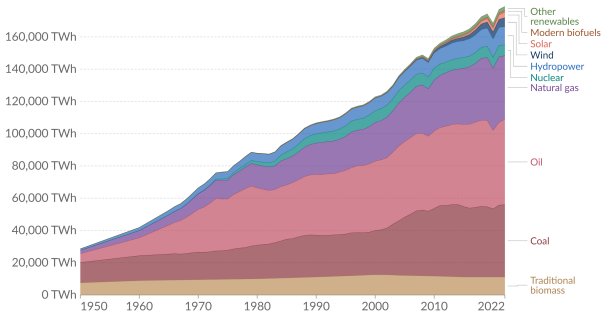
Why energy

Most of it is still fossil

Global primary energy consumption by source

Primary energy is calculated based on the 'substitution method' which takes account of the inefficiencies in fossil fuel production by converting non-fossil energy into the energy inputs required if they had the same conversion losses as fossil fuels.

Our World
in Data



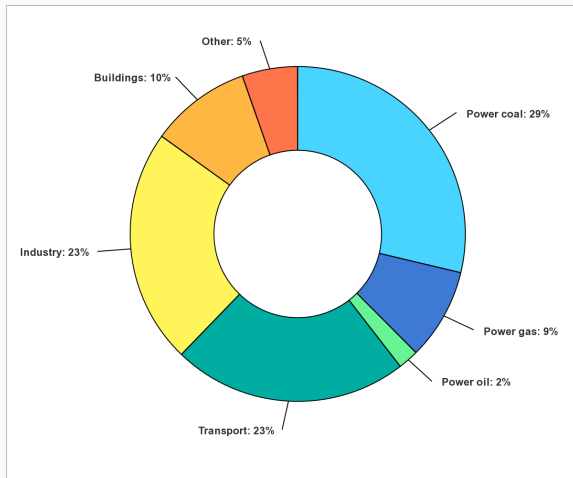
Source: Energy Institute Statistical Review of World Energy (2023); Vaclav Smil (2017)
OurWorldInData.org/energy • CC BY

Four fifths coal, oil and gas. Substituted before, never this fast, never on purpose.

Where the emissions come from

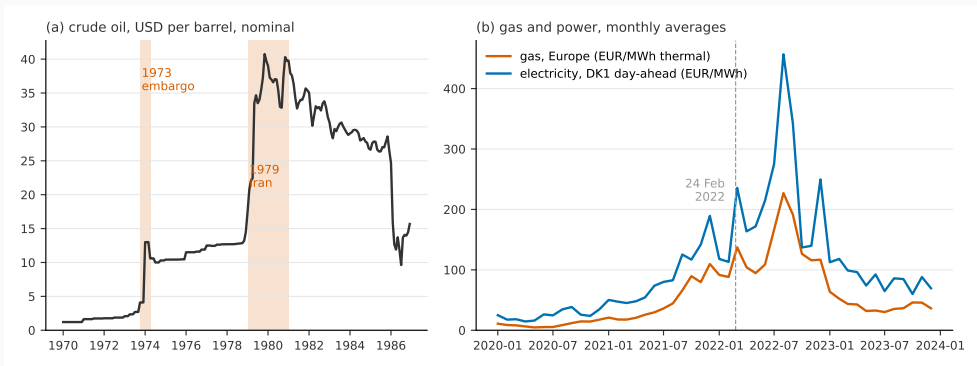
Of energy-related CO₂: **power** two fifths, **transport** and **industry** a quarter each, **buildings** a tenth.

Power comes first: cheapest substitutes, and electrification moves the other sectors' emissions into it.



Global energy-related CO₂ by sector. IEA, via Our World in Data.

Two energy crises, fifty years apart



World Bank Pink Sheet; Energi Data Service, DK1. Built by *figures/make_crises.py*.

Why did the electricity price follow gas, when most electricity was not made from gas?

Setup: one gas plant, two years

A combined-cycle gas plant. Three facts about it:

- **Efficiency** $\eta = 0.55$: one MWh of gas (thermal) becomes 0.55 MWh of electricity.
- **Emissions**: burning one MWh of gas emits 0.2 t of CO₂, and each tonne costs the carbon price τ .
- **Fuel**: gas is bought at the market price, per MWh thermal.

	2020	2022
Gas, EUR per MWh thermal	10	120
CO ₂ , EUR per tonne	25	80

In the hour we look at, wind and other cheaper plants cover 80% of demand, and this gas plant covers the rest.

The marginal cost of a gas plant, before and after

1. What does it cost this plant to produce one more MWh of electricity, in 2020 and in 2022?
2. What price does a wind farm receive for its MWh in that hour, and why?

Stop point

Two numbers in EUR/MWh, and one sentence beginning “the price is set by”.

If you finish early

A wind farm produces 1 TWh a year. What did 2022 do to its revenue?

Solution: the marginal plant sets the price

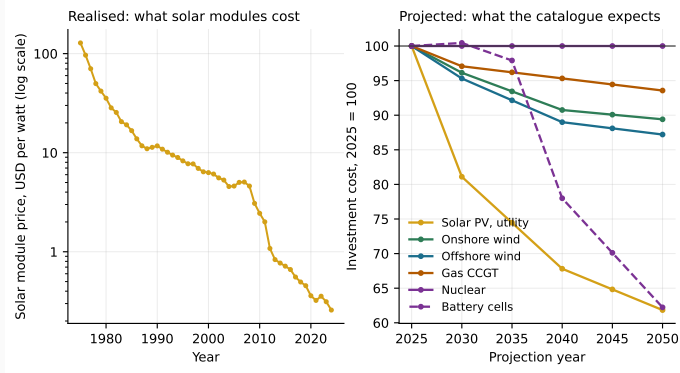
Marginal cost per MWh of electricity:

$$c = \frac{p^{\text{gas}} + \tau \varphi}{\eta} = \frac{p^{\text{gas}} + 0.2 \tau}{0.55}$$

	2020	2022
fuel, p^{gas}/η	18	218
carbon, $\tau\varphi/\eta$	9	29
marginal cost, EUR/MWh	27	247

- There is one price per hour, and every MWh sold in that hour gets it. It settles where the last plant needed to meet demand just covers its marginal cost.
- In 2022 that plant was, in most hours, a gas plant. So the gas price was the electricity price, for wind as much as for gas.
- The wind farm's costs did not change. Its revenue on 1 TWh went from about 27 to about 247 million euro. That is a **scarcity rent**, and it is the reason for the windfall taxes of 2022.

The cost collapse



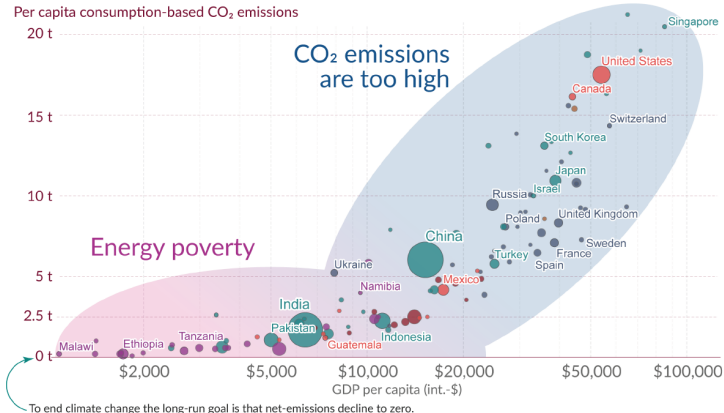
Generation Technologies, Fig. 5.1. Left: realised solar module prices, log scale. Right: what the cost catalogue expects.

Solar modules: 130 USD per watt in 1975, 0.26 in 2024. The cheap technologies exist. What is left is a **system-design question**.

The two energy problems

CO₂ emissions per capita vs GDP per capita

Our World
in Data



Data for 2017: Global Carbon Project, UN Population, and World Bank.

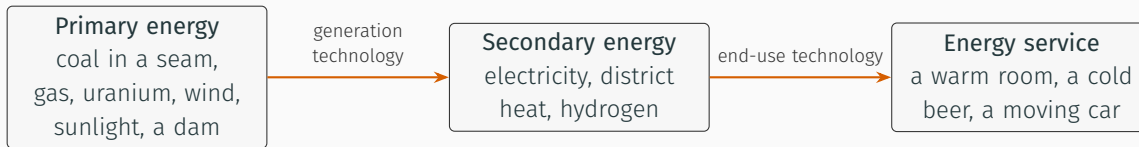
Five questions this course gives you the tools to answer

1. Does a renewable system need seasonal storage, and how much? lectures 4, 7
2. Do heat pumps help or hinder the integration of wind? lecture 5
3. Who gains and who loses from a cable to Norway, and who should pay for it? lecture 5
4. What sets the electricity price when nobody burns fuel? lecture 6
5. What does a carbon budget cost, and which policy delivers it cheapest? lectures 3, 7, 10

And in the second half: why the efficient fridge does not get bought, why adoption of a green technology feeds on itself, what negative emissions cost, and what energy has to do with geopolitics.

Not the **why** of the transition but the **how**, and how costly. Damages, political economy and leakage belong to other courses.

Energy economics 101



- Primary energy is a commodity and mostly behaves like one. Energy services are consumption and mostly behave like that.
- The middle step needs a large, long-lived, immovable, single-purpose piece of capital. **That is what makes energy markets peculiar.**
- A battery converts nothing. It appears anyway, because it competes with generation for the same job at the same times.

Generation Technologies, §1.1.

Power is not energy

- A **capacity** is a rate: kW, MW, GW.
- An **energy** is a rate times a duration: kWh, MWh, GWh.
- A 1 MW turbine running flat out for a year produces $1 \times 8,760 = 8,760$ MWh. A Danish onshore turbine actually produces about a third of that. The ratio is its **capacity factor**.
- Investment costs are quoted per unit of capacity (EUR/kW). Fuel costs and levelised costs per unit of energy (EUR/MWh).
- A plant that burns fuel consumes MWh_{th} and produces MWh_{e} , always fewer. The exchange rate is its efficiency.

Generation Technologies, §1.2 and Appendix C.

The primary energy fallacy

Useful energy out of 100 in	
Petrol car, at the wheels	20–30
Coal power plant, as electricity	35–40
Gas boiler, as heat	90
Electric car, at the wheels	80–90
Heat pump, as heat	300–400
Wind turbine, as electricity	100

Primary energy counts the fuel that went in. Two thirds of it is waste heat that nobody has to replace (Eyre 2021).

Electrification is efficiency.

How many offshore turbines to power Denmark?

Denmark uses about 35 TWh of electricity a year. A new offshore turbine has a capacity of 15 MW and a capacity factor of about one half.

1. How much energy does one turbine produce in a year?
2. How many turbines cover Danish consumption?
3. Danish demand peaks at about 6.5 GW on a winter evening. How does the fleet's capacity compare?

Stop point

A number of turbines, and one sentence on question 3.

If you finish early

The wind drops for a week in January. What now?

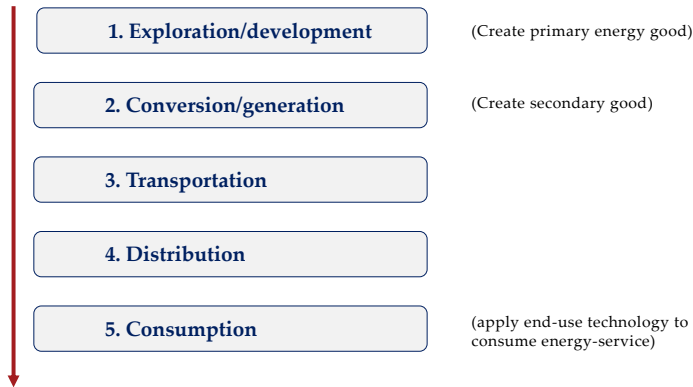
Solution: about 530 turbines, and that is not the problem

one turbine: $15 \text{ MW} \times 8,760 \text{ h} \times 0.5 \approx 66 \text{ GWh per year}$

the fleet: $35,000 \text{ GWh} / 66 \text{ GWh} \approx 530 \text{ turbines, about 8 GW}$

- Eight gigawatts of turbines against a peak load of six and a half: on average the fleet is more than enough. On a calm winter evening it produces nothing, and on a stormy Sunday it produces twice what anyone wants.
- The energy is cheap. What is hard is having it *when it is wanted, where it is wanted*: storage, interconnection, flexible demand, and something to run the other hours.
- Every one of those is a lecture. That is the balancing problem of the next slide.

A simplified supply chain



What the economist has to add

Coal and oil can be modelled with the tools you already have: a commodity, a world market, a price. From here on we have to care much more about

- **technology**: what each plant costs and can do,
- **geography**: from a national market to a region, a zone, a node,
- **short-run variation**: from a yearly model to an hourly one.

Predict before you see the table

Why do geography and short-run variation matter for electricity, heat and gas, and not for shoes?

Fill in the table for electricity, heat and gas

	Transport	Storage	Spatial distr.	<i>Transport</i> : can it be moved cheaply, by anything, without loss?
Coal	standard	standard	even	<i>Storage</i> : can it be kept cheaply, in anything, without loss?
Oil	standard	standard	concentrated	
Electricity	?	?	?	<i>Spatial distribution</i> : is the resource everywhere, or does someone have a natural monopoly?
Heat	?	?	?	
Gas	?	?	?	

Stop point

A filled table, and one sentence: which of the three is hardest to keep in balance, and why.

If you finish early

Hydrogen: which row does it resemble, and does that depend on distance?

Per unit of energy, which is the most expensive to store for a season?

1. Electricity
2. Heat
3. Natural gas

Electricity

	Transport	Storage	Spatial distr.
Electricity	networks	limited	idiosyncratic

- **Transport** needs a physical network built at large cost. Losses are low, but the wire has a capacity.
- **Storage** needs special technology: reservoirs, batteries, pumped hydro. Costly, and small relative to demand.
- **Spatial distribution:** secondary energy, so it can be made from anything; but hydro is in Norway, sun in Spain, wind in Denmark.

The conclusion

Supply must equal demand every second, with little storage and a capacity-limited network. That **balancing problem** is the subject of lectures 3 to 7.

Heat

	Transport	Storage	Spatial distr.
Heat	networks, local	accessible	idiosyncratic

- **Transport** needs a network, and losses grow with distance: district heat is a local market.
- **Storage** is comparatively cheap: a tank of hot water, a pit. It needs investment, but it works.
- **Spatial distribution:** again secondary energy; geothermal and solar heat are idiosyncratic, a boiler is not.

The conclusion

A balancing problem too, less acute because storage is accessible. And heat can be **electrified**, which ties the two markets together. That is lecture 5.

	Transport	Storage	Spatial distr.
Gas	mixed	mixed	semi-concentrated

- **Transport:** by pipeline within a few thousand kilometres, as LNG between continents through liquefaction terminals. Regional and global trade, on different infrastructure.
- **Storage:** underground, in depleted fields and salt caverns. Accessible, but a fixed installed capacity.
- **Spatial distribution:** concentrated enough for geopolitics.

The conclusion

Balanced over seasons rather than seconds. Because transport and storage are fixed capacities, a large supply shock produces a shortage rather than a price adjustment; 2022 was the example. And gas is an input to both electricity and heat.

The table, and what it says about the course

	Transport	Storage	Spatial distr.
Coal	standard	standard	even
Oil	standard	standard	concentrated
Electricity	networks	limited	idiosyncratic
Heat	networks, local	accessible	idiosyncratic
Gas	mixed	mixed	semi-concentrated

The first two rows are commodities and get standard tools. The last three are where the interesting economics is, because the physics of transport and storage forces the market to clear in time and in space. Five of the twelve lectures are about the electricity and heat rows.

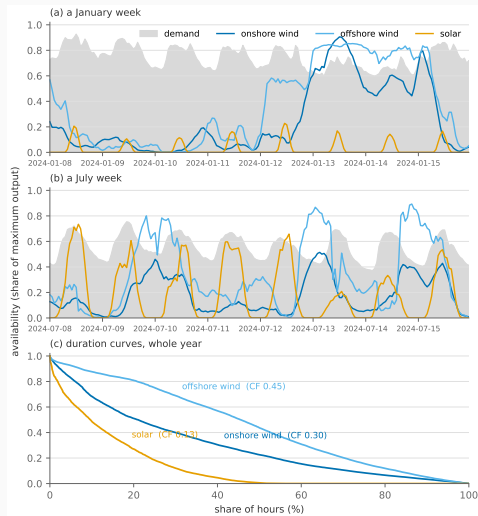
Hands on

What we will work with: the weather

Danish availability of wind and solar, hour by hour, 2019, and demand as the grey band.

- Wind comes in multi-day weather systems.
- Solar has a daily and a seasonal cycle, and is nearly absent in January.
- Capacity factors are far below one.

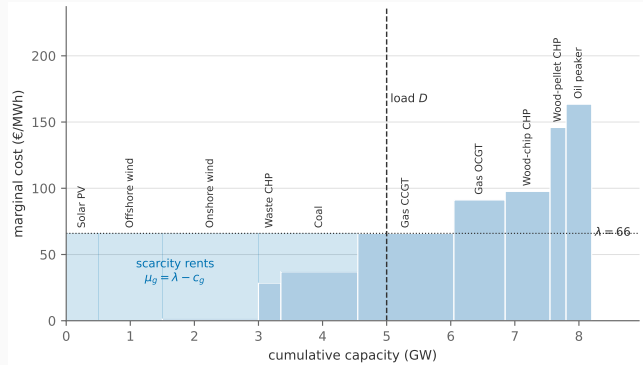
Energy System Models, Fig. 3.1. Lecture 3.



What we will work with: the merit order

Ten technologies in one hour, ordered by marginal cost. Load is the dashed line, the marginal plant sets the price, and the shaded area is the rent of every plant below it.

- The seed: every later lecture adds one component.
- The price is the shadow value of a constraint.

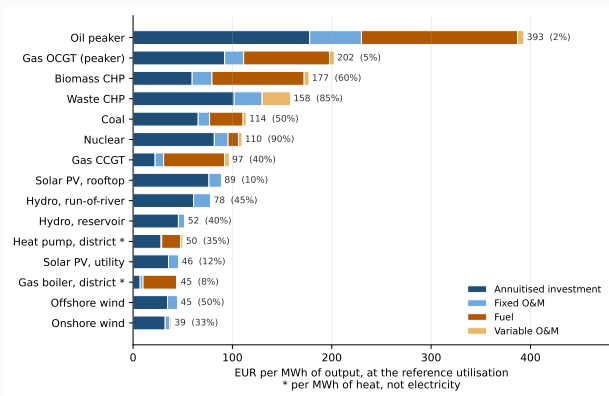


Energy System Models, Fig. 2.1.
Lecture 3.

What we will work with: the technologies

What a megawatt-hour costs from each plant, at its reference utilisation, split into capital, fixed, fuel and variable cost.

- Wind and solar are almost all capital.
- Peakers are almost all fuel, and run rarely.
- The single number hides the utilisation it assumes. The note spends a section on what it does and does not say.



Generation Technologies, Fig. 3.1.

The online materials

All the figures you see above live in the online repository – from data to model and simulations and figures.

So, while **the exam does not require that you know any programming or engage with any of the online materials**, I think it can be a great resource nonetheless for better learning. To this end, we will spend this last block on setting things up and check that they work.

Things to install/setup:

- Git/github desktop to clone repository.
- Python installation and environment setup.
- Github Copilot for students.
- Gurobi license for faster numerical optimisation (free student license).

The next 45 minutes

Three checkpoints, each with a stop point, on the next three slides. *INSTALL.md* in the repository is the guide; if you are already halfway through it, carry on.

1. **Run something.** Clone, create the environment, run *00-start-here.ipynb* to its first figure.
2. **Find something.** A term in the glossary, a number in a figure, a paper through the library.
3. **Break the assistant.** Completions off, one good question, then try to make it wrong.

Install not working?

Do 2 and 3 with the PDFs and notebooks read on github.com. Ask */debug-my-setup*, or me. Do not spend the evening on it; setup is not part of the course.

Checkpoint 1: run something

1. Clone: `git clone https://github.com/ChampionApe/EnergyEconomics.git`, or Code → Open with GitHub Desktop.
2. From the repository folder: `conda env create -f environment.yml`, then `conda activate EE2026`. Other Python setups are fine; read the dependencies out of `environment.yml`.
3. Check: `python -c "import pandas, pypsa, highspy; print('ok')"`
4. `jupyter lab`, open `00-start-here.ipynb`, run the cells until a figure appears.

Stop point

A merit-order figure on your screen, and the price it prints.

If you finish early

Raise demand in that cell until the price jumps. Which plant did you just switch on?

Checkpoint 2: find something

The material is designed to be looked up, not read cover to cover.

- *notes/technologies.pdf*, Appendix C is a glossary. Look up *capacity factor*, then *full-load hours*.
- *notes/energy-system-models.pdf*, Fig. 3.1(c): read the capacity factor of Danish offshore wind off the duration curve.
- *CURRICULUM.md*: find Hirth (2013), lecture 3. Open it through kb.dk with your KU login, not through the publisher.

Stop point

Two numbers (a capacity factor from the glossary's definition, and offshore wind's from the figure) and a PDF of Hirth open on your screen.

Checkpoint 3: break the assistant

1. Open the repository folder in VS Code with Copilot Chat, or in Claude Code. The assistant's commands appear in the chat's slash menu.
2. Click the Copilot icon in the status bar and *Disable Completions*. Chat stays on.
3. Ask `/model-explainer` what the market-clearing constraint in `model/dispatch.py` means economically.
4. Now try to make it wrong: ask it for a number it cannot know, ask it to solve an exercise for you, ask it about the exam.

Stop point

One sentence on what it did well and one on where it failed. If it failed, open an issue on the repository; it is new this year and I want to know.

- **Read** Mulder (2023), section 3, for the guest lecture.
- **Apply** for the GitHub Student Developer Pack today if you want Copilot; verification takes anything from minutes to days.
- **Finish** the install if it did not finish today, with */debug-my-setup* as first line of support.